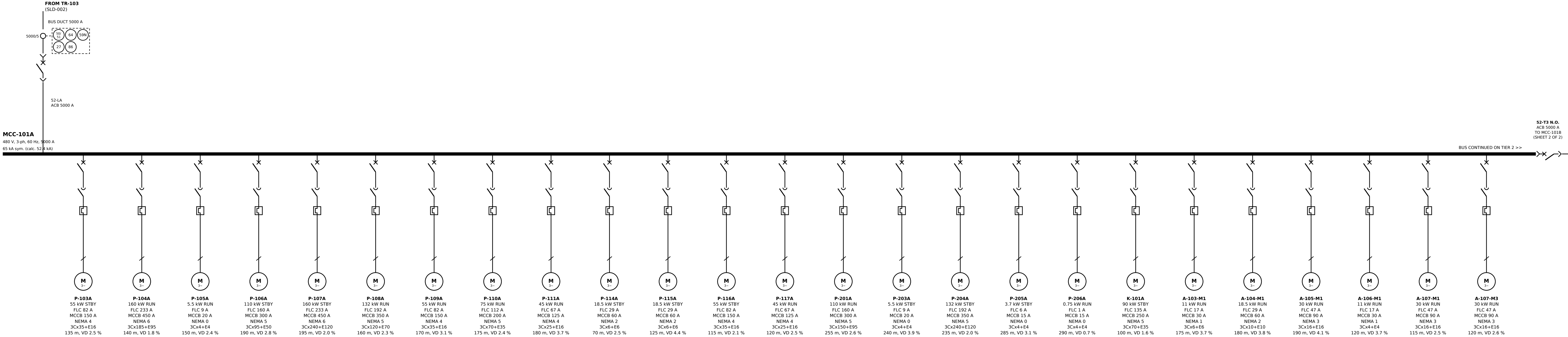


| MCC-101A SUMMARY                     | VALUE                                   |
|--------------------------------------|-----------------------------------------|
| Feeders / buckets                    | 50 (35 motor, 2 spare)                  |
| Connected load                       | 1,769 kW                                |
| Continuous / Interm. / Standby       | 1,084 / 127 / 559 kW                    |
| Maximum demand (C+0.3I+0.1S)         | 1,178 kW / 1,342 kVA, PF 0.88           |
| Both sections on one TR (tie closed) | 2,676 kVA vs TR 3000/3750 kVA ONAN/ONAF |
| Largest DOL motor start - bus dip    | 3.5 % (limit 10 %)                      |

- NOTES**
- MCC-101A: 480 V, 3-ph 3-W, 5000 A Cu bus, 65 kA sym.; ACB incomer/tie in LV switchgear section (IEEE C37.20.1), MCC sections UL 845, withdrawable buckets.
  - Motor buckets: MCCB (thermal-magnetic, <= 250 % FLC per NEC 430.52) + contactor + electronic OL, IEC 60947-4-1 Type 2 coordination; 120 V control transformer; DCS hardwired start/stop/run/trip.
  - Motors >= 30 kW; 4-20 mA motor current to DCS. VFDs: 6-pulse + passive filter (THDI <= 8 %).
  - Cable: Cu/XLPE/SWA/PVC 0.6/1 kV (IEC mm2). Sizes per CFU-000-EL-CBL-001 (ampacity x 1.25 FLC, VD <= 5 % running, <= 15 % at motor terminals during start, SC withstand via MCCB let-through).
  - Pump A/B pairs on opposite MCC sections; RUN/STBY = normally running unit for demand calculation.

| REV                                                                         | DATE       | DESCRIPTION              |
|-----------------------------------------------------------------------------|------------|--------------------------|
| A                                                                           | 2026-10-02 | Issued for review (FEED) |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |
| SINGLE LINE DIAGRAM<br>480 V SWITCHGEAR / MCC-101A                          |            |                          |
| DISCIPLINE                                                                  | STAGE      | SCALE                    |
| ELECTRICAL                                                                  | FEED       | NTS                      |
| SHEET                                                                       |            | 1 OF 2                   |
| DRAWN                                                                       | CHECKED    | APPROVED                 |
| CLAUDE CODE                                                                 | -          | -                        |
| DATE                                                                        |            | 2026-10-02               |
| DRAWING NUMBER                                                              |            | REV                      |
| CFU-000-EL-SLD-003                                                          |            | A                        |